

Analysis of Pizza Sales and Customer Ordering Behaviors

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Overview

Pizza is one of the most popular and widely consumed foods in the world. It is also one of my favorite foods, which made me especially interested in analyzing pizza sales data to see which types of pizza are most popular, which categories produce the most revenue, and how sales vary over time. This makes it an ideal topic for applying the data analysis techniques I learned in Professor Gupta's Information Data Science class.

The analysis will explore trends in pizza size and categories, revenue patterns, and peak ordering times throughout the day. Using descriptive statistics, visualizations, and aggregation techniques, the dataset will be analyzed to uncover key sales patterns and trends that could help pizza businesses make informed decisions about menu offerings, promotions, and inventory management.

Dataset Description

This report will use the "Pizza Sales Dataset" from Kaggle.com to perform the data analysis (<https://www.kaggle.com/code/mdismielhossenabir/pizza-sales-dataset/input>). For this project, I used a real-world pizza sales dataset containing detailed order records from a full year of operations in 2015. Although the data is historical, it provides a complete and accurate overview of pizza sales, making it an excellent dataset for applying analytical techniques learned in this course.

We have to look at what our dataset provides by first observing these properties:

1. **Columns, content, and relevance:** What information each column provides and how it can be used for analysis
2. **Dataset size:** The total number of rows and columns
3. **Data types:** Whether columns are numeric, text, or datetime, which changes analysis methods

The first thing I did was look at the first ten rows from the dataset to have a general understanding of kind of information the dataset provides. This can also be seen through `df.head`.

	A	B	C	D	E	F	G	H	I	J	K	L
1	pizza_id	order_id	pizza_name_id	quantity	order_date	order_time	unit_price	total_price	pizza_size	pizza_category	pizza_ingredients	pizza_name
2	1	1	hawaiian_m	1	1/1/2015	11:38:36	13.25	13.25	M	Classic	Sliced Ham, Pineapple, Mozzarella Cheese	The Hawaiian Pizza
3	2	2	classic_dlx_m	1	1/1/2015	11:57:40	16	16	M	Classic	Pepperoni, Mushrooms, Red Onions, Red Peppers, Bacon	The Classic Deluxe Pizza
4	3	2	five_cheese_l	1	1/1/2015	11:57:40	18.5	18.5	L	Veggie	Mozzarella Cheese, Provolone Cheese, Smoked Gouda Cheese, Romano Cheese, Blue Cheese, Garlic	The Five Cheese Pizza
5	4	2	ital_supr_l	1	1/1/2015	11:57:40	20.75	20.75	L	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, Green Olives, Garlic	The Italian Supreme Pizza
6	5	2	mexicana_m	1	1/1/2015	11:57:40	16	16	M	Veggie	Tomatoes, Red Peppers, Jalapeno Peppers, Red Onions, Cilantro, Corn, Chipotle Sauce, Garlic	The Mexicana Pizza
7	6	2	thai_chn_l	1	1/1/2015	11:57:40	20.75	20.75	L	Chicken	Chicken, Pineapple, Tomatoes, Red Peppers, Thai Sweet Chili Sauce	The Thai Chicken Pizza
8	7	3	ital_supr_m	1	1/1/2015	12:12:28	16.5	16.5	M	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, Green Olives, Garlic	The Italian Supreme Pizza
9	8	3	prsc_argla_l	1	1/1/2015	12:12:28	20.75	20.75	L	Supreme	Prosciutto di San Daniele, Arugula, Mozzarella Cheese	The Prosciutto and Arugula Pizza
10	9	4	ital_supr_m	1	1/1/2015	12:16:31	16.5	16.5	M	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, Green Olives, Garlic	The Italian Supreme Pizza
11	10	5	ital_supr_m	1	1/1/2015	12:21:30	16.5	16.5	M	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, Green Olives, Garlic	The Italian Supreme Pizza

(Source: from author)

Using `df.columns`, I see that the columns for this dataset included `pizza_id`, `order_id`, `pizza_name_id`, `quantity`, `order_date`, `order_time`, `unit_price`, `total_price`, `pizza_size`, `pizza_category`, `pizza_ingredients`, and `pizza_name`.

```
Index(['pizza_id', 'order_id', 'pizza_name_id', 'quantity', 'order_date',  
      'order_time', 'unit_price', 'total_price', 'pizza_size',  
      'pizza_category', 'pizza_ingredients', 'pizza_name'],  
      dtype='object')
```

(Source: from author)

From scanning the sample of the first ten rows of the data and looking at the column names with `df.columns`, I identified the columns that I thought would best used for my analysis to uncover key sales patterns and trends. The columns I decided were most relevant were `order_time`, `unit_price`, `total_price`, `pizza_size`, `pizza_category`, and `pizza_name`.

The next step is to check the size of the dataset and what datatypes is used for each column. I used `df.info()` to retrieve this information from the dataset. The size of the data set is 48,620 rows by 12 columns. This indicates that the dataset records a total of 48,620 pizzas sold in 2015, providing a snapshot of the company's sales for that year. We can also see from `df.info` that two columns are floats, three are integers, and seven are objects. Knowing the data types is important because it informs how we can analyze each column.

```
Data columns (total 12 columns):  
#   Column              Non-Null Count  Dtype  
---  -----  
0   pizza_id            48620 non-null  int64  
1   order_id            48620 non-null  int64  
2   pizza_name_id       48620 non-null  object  
3   quantity            48620 non-null  int64  
4   order_date          48620 non-null  object  
5   order_time          48620 non-null  object  
6   unit_price          48620 non-null  float64  
7   total_price         48620 non-null  float64  
8   pizza_size          48620 non-null  object  
9   pizza_category      48620 non-null  object  
10  pizza_ingredients    48620 non-null  object  
11  pizza_name           48620 non-null  object  
dtypes: float64(2), int64(3), object(7)
```

(Source: from author)

Prepare the Data

Before performing any analysis, the dataset must be check for issues such as missing values, incorrect data types, or inconsistent entries. Careful data preparation helps maintain accuracy in the analysis and leads to more accurate and dependable findings.

The first step is to check if our data contains any null values. We use `df.isnull().sum()` to see this.

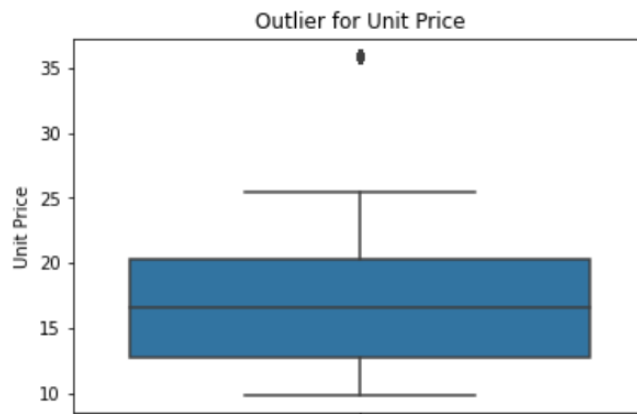
```
pizza_id      0
order_id      0
pizza_name_id  0
quantity      0
order_date    0
order_time    0
unit_price    0
total_price   0
pizza_size    0
pizza_category 0
pizza_ingredients 0
pizza_name    0
dtype: int64
```

(Source: from author)

From this, we can see that there are no null values in our dataset. We can also check for duplicates in the same manner using `df.duplicated().sum()`. The output for this is 0 which means there are no duplicates in our dataset.

After checking for missing values and duplicates, I reviewed the data types again using `df.info` to confirm that columns such as `order_date`, `unit_price`, and `total_price` were stored correctly.

I also inspected the `unit_price` column for potential outliers using a boxplot. The plot did show one unusually high price but I chose not to remove it. Pizza prices vary based on size and ingredients, so although the value was an outlier it was still a possible real sale rather than an error.



(Source: from author)

I also converted the `order_time` into numeric hours ranging from 0 to 23. This makes it possible to plot the data and analyze ordering patterns by hour. This is not possible in the current format `hour:date:time` (example: 11:38:36).

After reviewing each column, the dataset was found to be complete, consistent, and properly formatted. No additional cleaning was required before analysis.

Research Questions

Before analyzing the dataset, it is important to establish clear research questions. These questions will help us uncover key sales patterns and trends that could help pizza businesses make informed decisions.

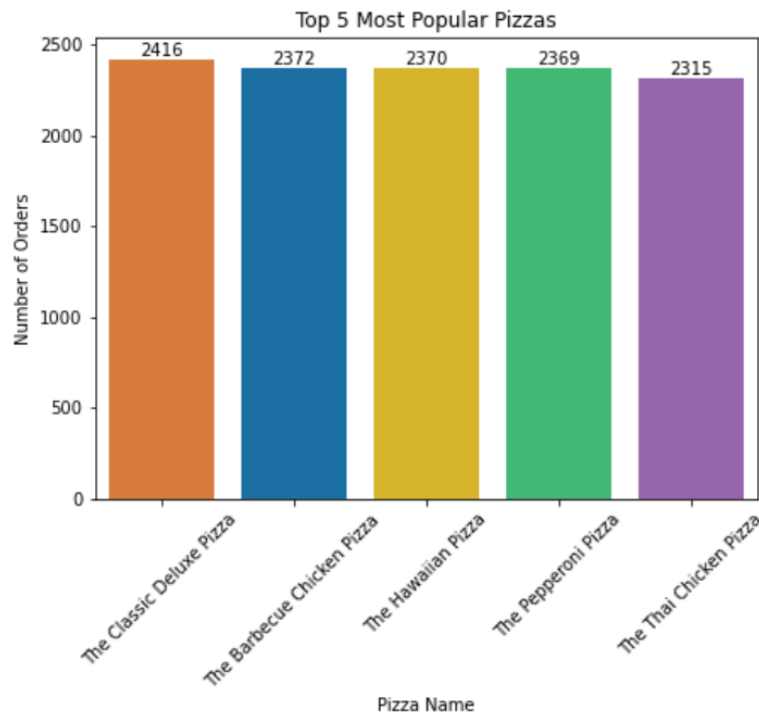
These six research questions will be addressed in the analysis:

1. What are the five most popular pizza varieties?
2. Which pizza size is ordered most frequently?
3. Which pizza categories generate the highest average revenue?
4. How do orders vary by time of day?
5. Is there a relationship between pizza size and total price?
6. How are total pizza order amounts distributed?

Analyze the Data

After preparing the data and asking research questions, I can start analyzing the data to answer these questions to help business owners identify trends and patterns.

1. What are the five most popular pizza varieties?

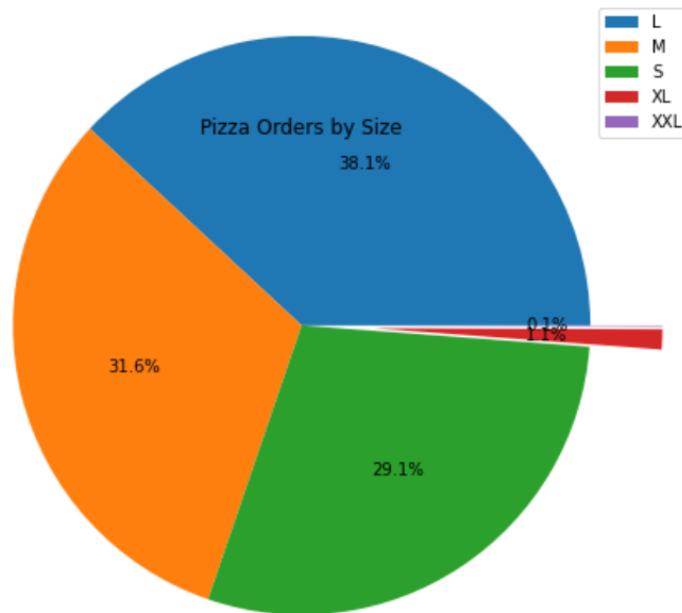


(Source: from author)

From the bar graph, we can clearly see that the five most popular pizza varieties are Classic Deluxe Pizza, Barbecue Chicken Pizza, Hawaiian Pizza, Pepperoni Pizza, and Thai Chicken Pizza. The number of orders for the pizza is really close, with a range from 2,416 to 2,315. This shows that the customer enjoys a variety of options rather than favoring just one. For a pizza

business, knowing which pizzas are consistently popular can help make sure ingredients are stocked and the menu meets what customers want.

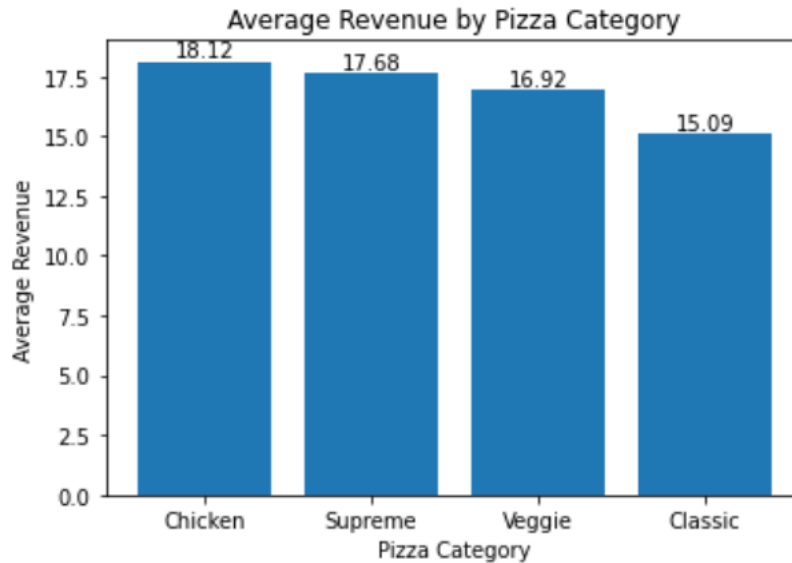
2. Which pizza size is ordered most frequently?



(Source: from author)

The pie chart shows that large, medium, small pizzas make up a majority of the orders, with large pizzas leading at 38.1%, medium pizzas at 31.6%, and small pizzas at 29.1%. Extra large pizzas and XXL pizzas are less common, at just 1.1% and 0.1% respectively. This suggests that customers generally stick to standard sizes, which can help pizza business plan inventory and prepare dough ahead of time to meet these demands.

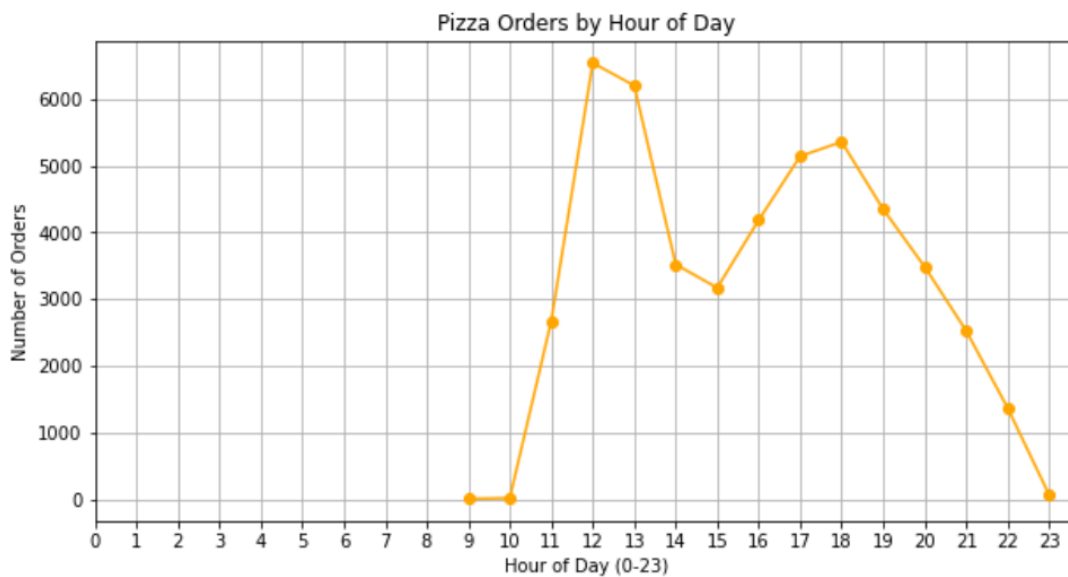
3. Which pizza categories generate the highest average revenue?



(Source: from author)

The bar chart shows the average revenue generated by each category. Chicken pizzas lead with an average revenue of \$18.12, followed by supreme at \$17.68, veggie at \$16.92, and classic at \$15.09. This pattern makes sense because pizzas with more meat or premium toppings generally cost more while simpler or vegetarian options tend to be less expensive. For a pizza business, this insight could help guide pricing strategies and menu planning to maximize revenue from higher priced categories.

4. How do orders vary by time of day?



(Source: from author)

The line chart shows how pizza orders vary throughout the day. We can see that the pizza business opens at 9 am and the order activity is low in the early morning. It rapidly rises around lunchtime and reaches its peak during the lunch hour. It then slowly declines until around dinner time, where it peaks once again, although not as high as at lunch. This daily pattern shows the importance of preparing lunch rushes, ensuring enough staff and ingredients are available at that time. It also brings the possibility of running promotions or special deals to encourage orders during slower hours.

5. Is there a relationship between pizza size and total price?



```

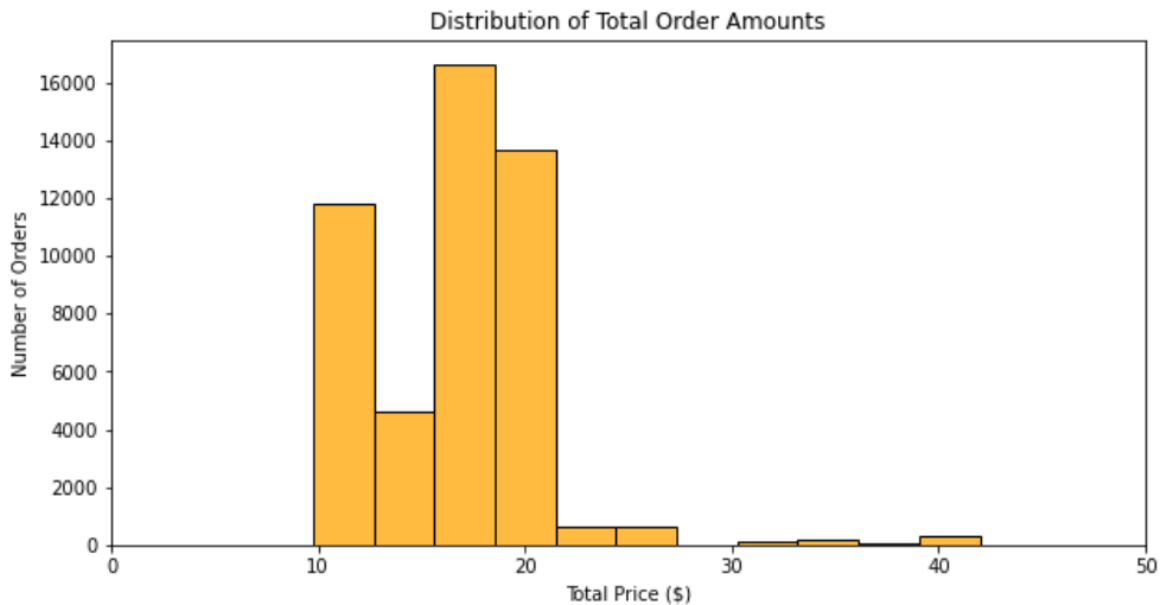
pizza_size
S      12.364328
M      15.951219
L      19.802553
XL     25.500000
XXL    35.950000
Name: unit_price, dtype: float64

```

(Source: from author)

The scatterplot and average unit price show a relationship between pizza size and total price. While the points are not perfectly aligned, the overall trend of larger pizzas costing more can be seen. On average, small pizzas are about \$12.36, mediums are \$15.95, larges are \$19.80, XL are \$25.5, and XXL are \$35.95. This shows a positive correlation where if size increases, the price tends to increase as well. Individual orders can slightly vary due to expensive toppings or ingredients.

6. How are total pizza order amounts distributed?



(Source: from author)

The histogram shows how total pizza order amounts are distributed. Most orders fall in the price range between \$10 and \$20 which means customers usually spend within that range when they buy pizza. There are no orders below \$10 which indicate this pizza business does not sell pizza under \$10. The number of orders fall dramatically pass \$20 meaning that the more expensive pizzas are rarely ordered. This graph suggests that most customers tend to place smaller or average sized orders rather than large higher priced ones.

CONCLUSION

This analysis gives a clear picture of customer behavior and sales patterns in the pizza dataset as it highlights several insights that can help businesses make better decisions. One thing I learned is that the top pizza varieties are extremely close in popularity, which shows that customers enjoy a variety of options rather than one dominant choice. The business should continue offering variety rather than one single best seller. With small, medium, and large pizzas making up majority of the orders, the business should prioritize inventory and prep work around these standard sizes.

The revenue analysis shows a clear trend as well. The pizzas with meat and more toppings have the highest average revenue due to ingredients and pricing. This can be useful for the business for future marketing efforts if the business wants to increase profitability. The time-of-day chart is useful because it shows the peaks for lunch and dinner which helps identify when business should increase staff or time-based promotions. The relationship between pizza and price also made sense with the general trend showing larger sizes bring in more revenue. Finally, the

histogram of total order amounts showed that most customer spend between \$10 and \$20 which can be useful for creating deals or bundles.

For future work, it would be helpful to analyze customer behavior across seasons. Our current analysis gives an estimated activity for each day on average in a year. However, the actual amount can vary depending on season. It could also be useful to explore other varieties and how deals can affect sales. Adding demographic or location-based information could also reveal patterns that a yearly pizza dataset cannot capture. Overall, the analysis could answer the research questions but it also leaves room for future questions that could benefit the pizza business even more.

REFERENCES

Code used:

Attached with the report

Dataset used:

<https://www.kaggle.com/code/mdismielhossenabir/pizza-sales-dataset/input>

External links used:

Color palette: https://seaborn.pydata.org/tutorial/color_palettes.html

Line graph: <https://www.geeksforgeeks.org/python/plot-line-graph-from-numpy-array/>

Histogram: <https://medium.com/@whyamit404/what-is-numpy-hist-and-how-does-it-work-65ebd55574cc>

Graphs: https://www.tutorialspoint.com/numpy/numpy_matplotlib.htm